

SingleRAN

Inventory Management Feature Parameter Description

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<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for board storage resetting by the MEHC. For details, see 3.3.3 Hardware .	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
SRA N	MRFD-210303	Inventory Management	3 Inventory Management
LTE FDD	LBFD-004013	Inventory Management	
LTE TDD	TDLBFD-0040 13	Inventory Management	
NB- IoT	MLBFD-12000 413	Inventory Management	
NR	FBFD-010025	Inventory Management in Basic O&M Package	

3 Inventory Management

3.1 Principles

3.1.1 Introduction

Inventory management is performed on the MAE. With this feature, the MAE obtains required information about the physical and logical resources of managed network elements (NEs) through various interfaces. The information helps users comprehensively manage physical resources (such as equipment) and logical resources. For the definition of inventory management, see 3GPP TS 32.690 V7.0.0 (2007-06).

Inventory management objects include:

- Physical resources
Cabinets, subracks, slots, boards, ports, remote electrical tilt (RET) antennas or Remote Global Positioning System (RGPS)
- Logical resources
Versions

The inventory data includes:

- The static data about hardware and firmware units (for example, cable clips, processing units, or power supplies) for constructing the NEs managed by the OSS. The data has the following characteristics:
 - Usually provided by vendors.
 - Independent from the operating status of the related equipment or units.
 - Seldom changed during normal operation.
 - Cannot be modified through the northbound interface.
 - Independent from configuration management.
- The data that can be integrated in or assigned to related equipment or units during installation or operation.
- The data that contains the physical relationships between equipment or units. For example, board *A* is installed in slot *B*.

3.1.2 Principles

Inventory management is performed at three layers:

- NE layer: NEs collect inventory data and report the data as the MAE requires.
 - The traditional base station reports an inventory device file to the MAE.
- Element management system (EMS) layer: The MAE collects the inventory data of all managed NEs, provides the data to the NMS, and implements basic management functions such as querying or viewing the inventory data.
- NMS layer: The NMS provides operators with the application of the inventory data, such as stock and spare parts management.

Figure 3-1 shows the principles of inventory management.

Figure 3-1 Principles of inventory management

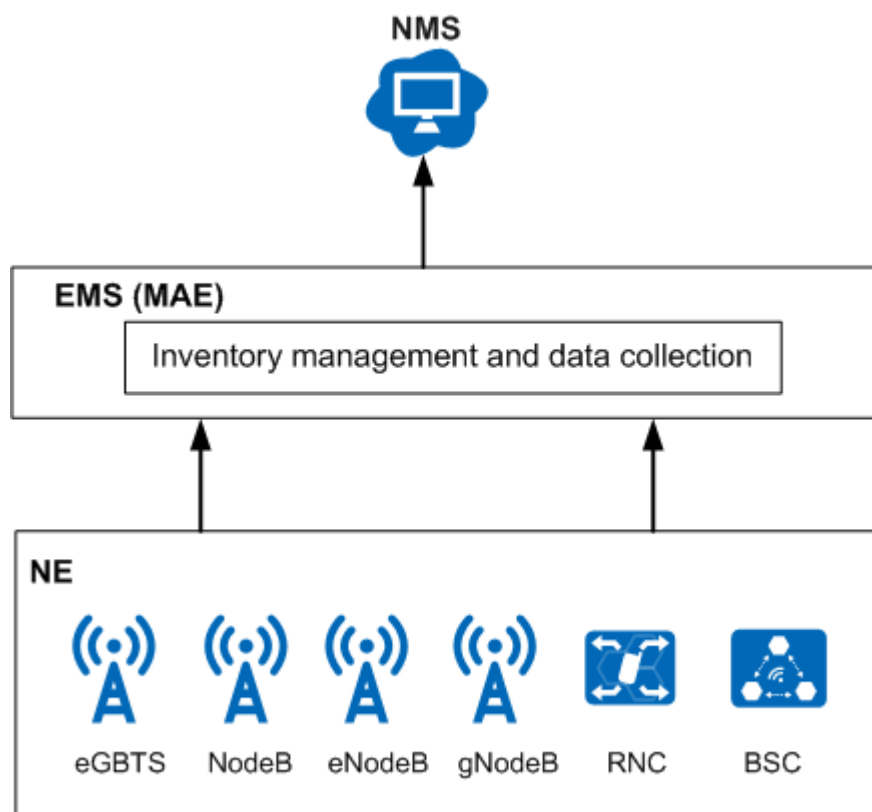


Figure 3-2 and Figure 3-3 show the inventory management processes for physical and logical resources.

Figure 3-2 Inventory management processes for physical resources

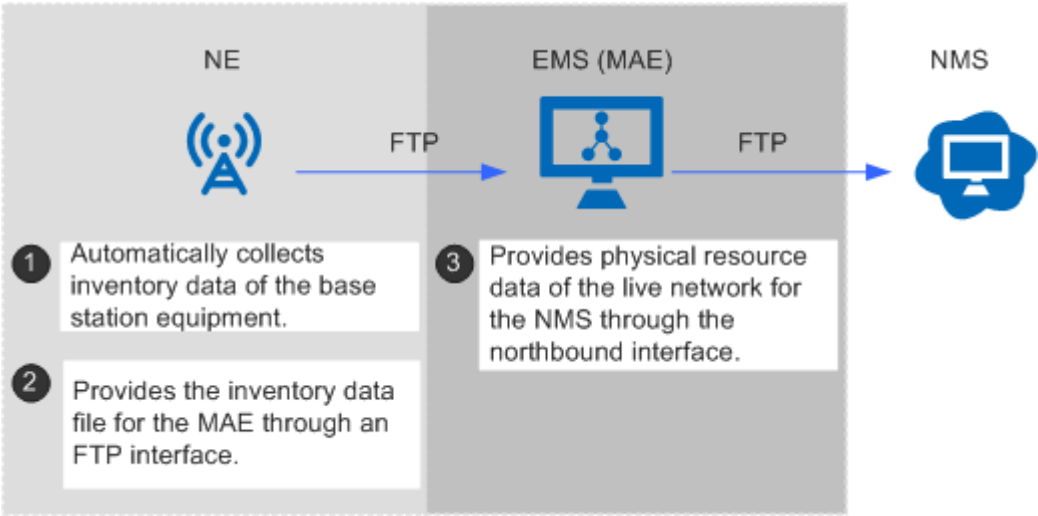
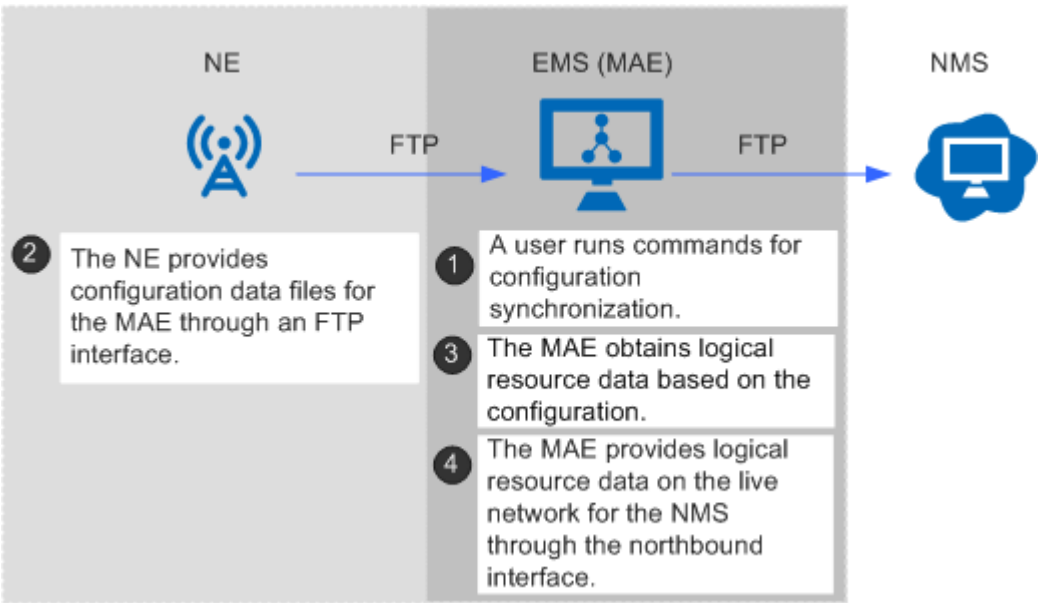


Figure 3-3 Inventory management processes for logical resources



The information about physical and logical resources is included in one XML file and sent to the NMS.

3.1.3 Functions

3.1.3.1 Importing Inventory Data

This function enables the MAE to import inventory data of NEs.

Table 3-1 Importing inventory data

If...	Then...
NEs can report their inventory data to the MAE	The MAE imports inventory data to ensure data consistency between the MAE and NEs using the method specified in 3.1.3.1.1 Synchronizing Inventory Data .
NEs cannot report their inventory data to the MAE	Users manually add inventory data to the MAE using the method specified in 3.1.3.1.2 Adding Inventory Data .

3.1.3.1.1 Synchronizing Inventory Data

Synchronization on a Scheduled Basis

The MAE can synchronize the inventory data of NEs on a scheduled basis.

1. Users set regular tasks on the MAE, and the MAE regularly delivers a command to NEs for synchronizing inventory data.
2. After receiving the command, NEs export inventory data and upload the data to the MAE.
3. The MAE stores the data.

Manual Synchronization by Users

When users need to view the inventory data immediately after new equipment is installed, they can manually synchronize the inventory data of the NEs to the MAE. The MAE can synchronize the inventory data with multiple NEs at the same time.

Synchronization Triggered by Inventory Data Changes

When the inventory data of an NE changes (for example, a board on the NE is replaced), the NE reports an inventory data change notification to the MAE. After receiving the notification, the MAE initiates inventory data synchronization. The minimum synchronization interval is 1 minute.

3.1.3.1.2 Adding Inventory Data

Manually Adding Inventory Data by Piece

If the number of devices whose inventory data needs to be added is small, users can manually add the data of each piece of equipment to the MAE database.

Manually Adding Inventory Data in Batches

Users can add inventory data to the MAE in batches by importing inventory data files.

3.1.3.2 Querying Inventory Data

This function enables users to view inventory data of resources on the live network. After users set query conditions, the MAE lists all of the inventory data that meets the conditions in the network.

3.1.3.3 Modifying Inventory Data

This function enables the MAE to modify inventory data of NEs.

Table 3-2 Modifiable fields

Data Source	Field
Synchronized inventory data	• Rack • Frame • Slot • Data of Last Service under the Port node • Data of Last Service under the Board node
New inventory data	The values of key attribute fields that are marked with an asterisk (*) cannot be modified. The values of other fields can be modified.

3.1.3.4 Deleting Inventory Data

The synchronized inventory data cannot be manually deleted from the MAE. If a piece of equipment is removed from an NE, the inventory data file synchronized to the MAE will not include the inventory data of this piece of equipment and the MAE automatically deletes the inventory data of this equipment from the database.

Users can manually delete the new inventory data from the MAE. After the inventory data of an equipment node is deleted, the inventory data of its child nodes is also deleted.

3.1.3.5 Exporting Inventory Data

The inventory data in the database of the MAE can be exported to an XML, TXT, or CSV file and saved on a local disk for view. Users can export the inventory data of all MOs under an equipment node or that of a specific MO.

Exporting Inventory Data by Piece

The MAE can export all or partial inventory data of an NE, the inventory data of a single MO, or the inventory data that is manually imported.

Exporting Inventory Data in Batches

The MAE filters inventory data based on NEs or MOs, and then exports the filtered data to a file.

3.1.3.6 Electronic Label

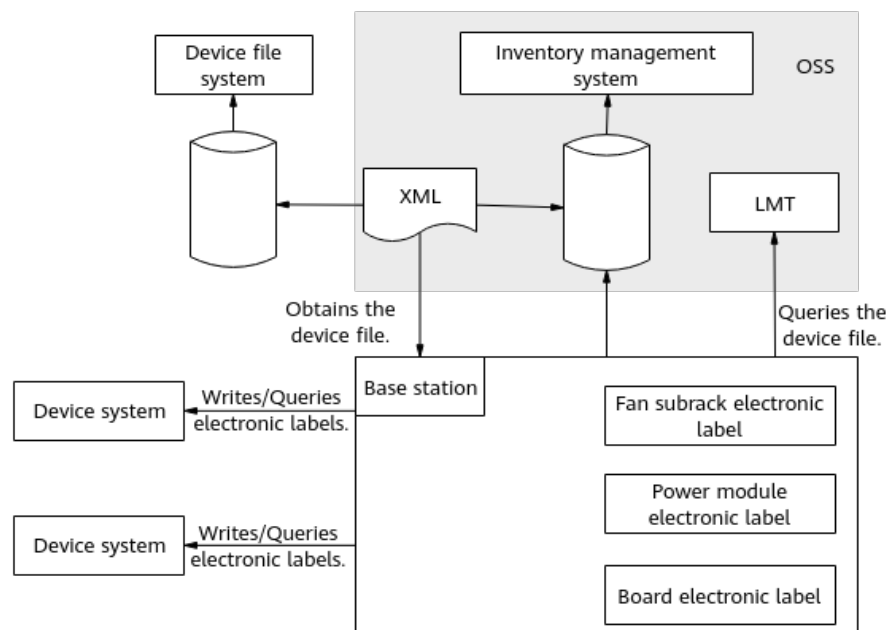
An electronic label uses electronic data to identify the object code. It applies to any hardware equipment produced or purchased by Huawei. The inventory management system automatically queries electronic labels when requiring an NE to export the device file. The query results are filled in the corresponding positions in an XML file.

Huawei electronic labels include the following items:

- BoardType
- BarCode
- Item
- Description
- Manufactured
- VendorName (Huawei)
- IssueNumber
- CLEICode
- BOM
- RET system

Figure 3-4 shows the interaction between electronic labels and the inventory management system.

Figure 3-4 Interaction between electronic labels and the inventory management system (a base station used as an example)



- The inventory management system automatically queries electronic labels when requiring an NE to export the device file. The queried electronic labels are filled in the corresponding positions in XML files.
 - Using the inventory management system: After querying the electronic label of a board with optical modules, the inventory system traverses all

optical modules on the board based on port numbers and obtains their electronic labels. For example, a GBSC delivers a command to query the electronic labels of optical modules in a GBTS. The GBTS reports the electronic labels of its optical modules to the GBSC. After receiving these electronic labels, the GBSC saves them in an inventory table and reports the table to the MAE.

- Manual query: The maintenance engineer runs the following command on the OSS to query the electronic labels of optical modules.
 - GBTS: **DSP BTSELABEL**
 - eGBTS/NodeB/eNodeB/gNodeB: **DSP ELABEL**
- Synchronized inventory information including electronic label information is stored in the inventory database of the MAE.

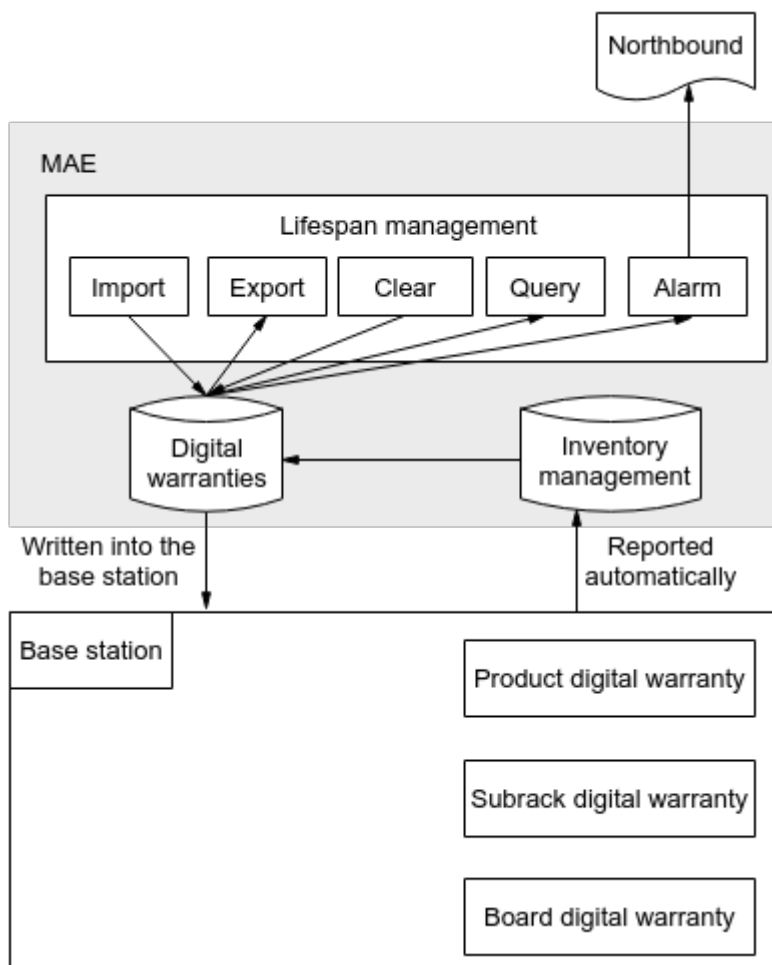
3.1.3.7 Digital Warranty

A digital warranty contains information such as the lifespan information of all hardware equipment produced or purchased by Huawei. The query results of the lifespan information in a Huawei digital warranty include the following items:

- NE
- Category
- Unit Type
- Unit Name
- Serial Number
- Manufacture Date
- Starting Point Date
- Session
- Expiry Date
- Status

Figure 3-5 shows the interaction between the lifespan management and the inventory management.

Figure 3-5 Interaction between the lifespan management and the inventory management



- The lifespan information query function of digital warranties
 - MAE query: The query results of the lifespan information include NE, Category, Unit Type, Unit Name, Serial Number, Manufacture Date, Starting Point Date, Session, and Status. These are mandatory fields that are displayed on the customer interface.
 - MAE alarm: The lifespan alarm switch is configurable and is turned off by default. When the switch is turned on, the periodic inspection is started as scheduled. Alarms are reported once the reporting conditions are met, and cleared once the clearance conditions are met. When the switch is turned off, lifespan-related alarms are cleared on the alarm console. The setting of this switch does not affect the import, export, and query of the lifespan information.
 - Manual query: The maintenance engineer runs the **DSP EGUARANTEE** command on the MAE or the LMT to query the Starting Point Date and Session of a board or a product. The information is synchronized from the MAE according to the imported digital warranty data.
- The lifespan information maintenance function of digital warranties

- MAE for import: The files to be imported must be in the CSV format. After the lifespan information is imported, the system displays the number of valid lifespan information records.
- MAE for export: The exported files are in the CSV format. The following information can be exported:
 - Lists of all products, boards, and parts managed by the OSS, including objects whose lifespan information is complete and those whose information is missing. The exported list is usually used for O&M contract renewal.
 - Lists of all products, boards, and parts managed by the OSS whose Starting Point Date and Session fields in the lifespan information are empty. The list is consistent with that exported based on the Lack of the lifespan information alarm. The exported list is usually used to add the lifespan information.
- MAE for clearance: If you find that the imported lifespan information is incorrect or you want to delete this information, you can use this function to clear the digital warranty data. This is a high-risk operation. Exercise caution when performing this operation. If a misoperation is performed, the deleted digital warranty data cannot be restored and needs to be imported again.

 NOTE

It is recommended that the value of Session in digital warranties of wireless products be less than or equal to **240** when the information is imported to the MAE. Generally, the period specified in the contract does not exceed 120 months.

Digital warranties of major hardware devices such as the main control boards, baseband processing units (BBPs), and RF modules in a base station can be queried using the **DSP EGUARANTEE** command while those of other hardware devices in a base station cannot be queried using this command. To query the digital warranty data of all boards in a base station, you are advised to use the related query function of the MAE.

3.1.3.8 Resetting Board Storage

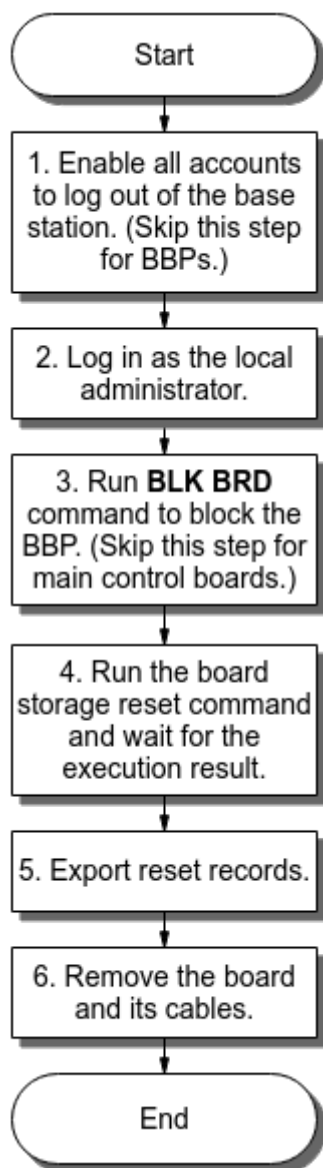
The **RST BRDSTO** command can be used to erase all user data for main control boards, BBPs, multi-mode extended radio frequency cards, and multifunctional extended digital cards. This reduces the risk of user data leakage.

 NOTE

Typical user data includes service configuration data, account names, and passwords.

Figure 3-6 shows the detailed procedure for this function.

Figure 3-6 Procedure for resetting board storage



Step 1 Enable all accounts to log out of the base station. (Skip this step for BBPs.)

Step 2 Log in as the local administrator.

Step 3 Run the **BLK BRD** command to block the BBP before resetting it. (Skip this step for main control boards.)

Step 4 Run the board storage reset command and wait for the execution result.

Step 5 Export reset records.

This function provides two types of reset record to save related reset credentials for users: MML command execution results and reset logs. For details, see [Table 3-3](#).

Table 3-3 Reset records

Reset Record	Content of the Reset Record	
MML command execution results	The MML command execution results contain key information about all data reset records. For details, see <i>MML Command Reference in 3900 & 5900 Series Base Station Product Documentation</i> .	
Reset record logs	Operation logs	After an MML command is executed successfully, the operation log will be updated. • Level 1: Two operation records are generated, indicating a successful execution and the execution result. • Level 2: – For BBPs, two operation records are generated, indicating a successful execution and the execution result. – For main control boards, only one operation record is generated, indicating the execution result.
	Security logs	After an MML command is executed, security logs are recorded. The log content is the MML command execution result.

For details about MML command execution results contained in the logs, see *MML Command Reference in 3900 & 5900 Series Base Station Product Documentation*. You can use the Notepad to open the logs or contact Huawei engineers and use a log tool to view the logs.

Step 6 Remove the board and its cables.

----End

 CAUTION

- Read the precautions in the MML command reference carefully and select a reset level as required. A higher level indicates a larger range of data to be deleted.
- If resetting board storage is required for both the main control boards and BBPs, reset the BBPs and then the main control boards.
- If other MML commands are executed after board storage resetting is complete, an exception may occur and data may not be completely erased. To prevent such an exception, upload the operation logs and security logs to a local PC, and then remove the board as soon as possible.

When resetting the storage of a main control board, pay attention to the following:

- Running **RST BRDSTO** will permanently delete user data, and such data cannot be restored. Therefore, exercise caution when running this command, and back up the required data before running this command. For details about how to back up data, see **Maintaining the Case-Structured BBU > Replacing BBU Components > Replacing the UMPT > Back up the configuration files and the license** in *BBU Hardware Maintenance Guide in 3900 & 5900 Series Base Station Product Documentation*.
- After the command is executed, configurations are prohibited. Therefore, set the FTP connection in advance to ensure that logs can be exported. To retain the MML command output, do not reset the base station after command execution.
- If a file to be deleted using the **RST BRDSTO** command is locked by other tasks, it cannot be deleted. In this case, you are advised to reset the base station and run the command again. If the base station is manually reset after the MML command execution fails, or the base station experiences a power-off reset or the board is removed and then inserted during the MML command execution, the IP address may be restored to the default one or the account and password may be restored to the default ones after a board restart because some files may have been deleted.
 - If you cannot log in to the base station using the original IP address, log in to the base station using the default IP address.
 - If the original account and password cannot be used to log in to the base station, use the default administrator account and password to log in to the base station. After logging in to the base station again, run the MML command again.

3.2 Network Analysis

3.2.1 Benefits

Table 3-4 Benefits provided by inventory management

Application Scenario	Description	Benefit
Network deployment	Users collect information about installed equipment in a network and check whether all the ordered equipment has arrived at the site.	O&M costs are reduced.
Routine network O&M	• Users obtain the physical configurations of network equipment and check logical resource configurations such as the software and patch versions of the current equipment. • Before the serving period of network equipment ends due to the equipment aging and deterioration during network operation, inventory management triggers alarm reporting to inform users of equipment replacement. The MAE currently does not support this function.	O&M operations are simplified, reducing O&M costs.
Network troubleshooting	• If equipment is faulty and must be replaced, inventory management helps maintenance personnel obtain specifications of backup equipment and other inventory information such as the storing position. • Inventory management also helps users obtain accurate data for making service decisions in a timely manner. For example, locate other boards of the same type and produced at the same date as the faulty board.	Faults are quickly rectified.
Network upgrade and capacity expansion	Users collect information about new equipment (such as cable clips, processing units, or power supplies).	Equipment data is quickly measured.

3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

This function has no impact on the network.

3.3 Requirements

3.3.1 Licenses

None

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

None

3.3.2.2 Mutually Exclusive Functions

None

3.3.2.3 Function Impacts

None

3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

All GSM base station models

All UMTS base station models

All FDD base station models

For NB-IoT, the following base stations are compatible:

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

All TDD base station models

All NR base station models

Boards

Board storage resetting is supported only by the following boards:

- UMPT series
- UBBP series
- MDUC
- UMDU
- MERC
- MEDC
- MEHC

NOTE

The integrated BBU integrates the main control, transmission, and baseband modules. In an integrated BBU3910A/BookBBU5901, the UMDU is the logical board that supports the main control function. In an integrated BBU3910C/BBU5900C, the MDUC is the logical board that supports the main control function.

RF Modules

None

3.3.4 Others

None

3.4 Operation and Maintenance

The MAE provides inventory management functions. For details, see "Inventory Management" in *MAE Product Documentation*.

To start inventory management, choose **Configuration > Inventory Management** on the MAE.

NOTE

The function of resetting board storage is invisible on the MAE.

4 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

5 Reference Documents

1. 3GPP TS 32.690 V7.0.0 (2007-06): "4 Inventory Management (IM) concepts"
2. 3GPP TS 32.692 V7.0.0 (2007-06): "6.5 Information attribute definitions"
3. *MAE Product Documentation*
4. *MML Command Reference in 3900 & 5900 Series Base Station Product Documentation*
5. *BBU Hardware Maintenance Guide in 3900 & 5900 Series Base Station Product Documentation*